Tutorial 5 – Software Structure and Architecture

1. What are architectural views, and why they are important in software architecture?

2. There is a say that it is useful to describe architecture through different viewpoints and views. What do you think about this? Defend your answer.

3. You have been asked to develop an online retail system that sells groceries. You have decided to use the 4 + 1 view model of software architecture to design the system. Describe each of the components, together with a relevant example in the 4 + 1 view model.

4. What is an architectural style/pattern?

5. Is software architecture restricted to only one architectural style/pattern for its logical design, or can they include more than one architectural style/pattern? Explain with examples.

6. Explain the following architectural styles/patterns, and provide an example of a system appropriate for them. Explain how these styles/patterns support particular quality attributes.
   (a) Blackboard
   (b) Pipe and filter
   (c) MVC
   (d) Layered
   (e) Broker

7. Compare and contrast the following architectural styles/patterns:
   (a) Client-server vs. Broker
   (b) Layered vs. Main program and subroutine

8. How can the pipe and filter architectural style/pattern enhance the performance of software systems?

9. What is the fundamental difference between a fat-client and a thin-client approach to client–server systems architectures?

10. Your customer wants to develop a system for stock information where dealers can access information about companies and evaluate various investment scenarios using a simulation system. Each dealer uses this simulation in a different way, according to his or her experience and the type of stocks in question. Suggest a client–server architecture for this system that shows where functionality is located. Justify the client–server system model that you have chosen.

11. When would you recommend the use of a multi-tier client-server architecture?

12. You have been asked to design a secure system that requires strong authentication and authorization. The system must be designed so that communications between parts of the system cannot intercepted and read by an attacker. With an aid of a diagram, suggest the most appropriate client-server architecture for this system, giving reasons for your answer.

13. Explain the roles of objects of each class for the Model–View–Controller software design architecture.

14. In the design of software architecture, *layering* and *partitioning* are two important approaches that will be used by most software engineers.

    Differentiate between *partitioning* and *layering* in the system development context and explain why system developers prefer to divide a software system into subsystems with different layers.

15. 2024 May/June (Past Exam)

Question 2

a) List the classification of architectural patterns. (5 marks)

b) Robert is the new software architect for an internal IT team of a car assembly factory. The factory assembles a new car, from preparing the base and the car's frame until a complete unit is produced at the end of the production line. The new unit will move from one (1) station to the other using an automated conveyor. Each station will fit different components on the new unit. All assembly stations are automated and system-controlled. The factory has instructed Robert and his team to develop a new system, replacing the current system to manage, control, and monitor the stations.

   (i) Which architectural pattern do you think Robert will implement in this project? Provide **TWO (2)** justifications to support your answer. (6 marks)

   (ii) Construct the architectural diagram selected in Question 2 b) (i) using the following description. (10 marks)

   After the car frame is installed, the new unit will move to the second station to have its engine installed. Once complete, it moves to the third station, called the hood station. At this station, the unit will be fitted with a hood. Then, the wheels will be installed on the new unit at the fourth station.

c) Explain **TWO (2)** views from the UML 4+1 Views relevant to programmers. (4 marks)

[Total: 25 marks]

Additional Questions (Home Work)

a) A company is developing an online virtual banking system which provides banking services to the customers. The system is proposed to “virtually” represent a bank and providing almost 90% of bank services. The system requires an interaction between system components where the effects on each component is not significant. Components have their own functions and works either within the group of components with similar functions or different functions.

(i) Analyze and compare **THREE (3)** architecture patterns that are suitable for the above scenario. These architectures may appropriate either independently for the system or hybrid architectures. You may draw a table for the comparison purpose.

(ii) Based on the architecture patterns suggested from **Question 1.(i)**, demonstrate and explain how the architecture of the online virtual banking is. You are required to draw a diagram to demonstrate the architecture with the pattern you have selected. In your diagram, you are required to include the examples of components that lay under different parts of the architecture pattern. You can apply more than one pattern for the design.